In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Create a sample diabetes dataset
data = {
 'Pregnancies': [6, 1, 8, 1, 0, 5, 3, 10, 2, 4],
 'Glucose': [148, 85, 183, 89, 137, 116, 78, 115, 197, 110],  'BloodPressure': [72, 66, 64, 66, 40, 74, 50, 0, 70, 92],
 'SkinThickness': [35, 29, 0, 23, 35, 0, 32, 0, 45, 0],
 'Insulin': [0, 0, 0, 94, 168, 0, 88, 0, 543, 0],
 'BMI': [33.6, 26.6, 23.3, 28.1, 43.1, 25.6, 31.0, 35.3, 30.5, 37.6],
'DiabetesPedigreeFunction': [0.627, 0.351, 0.672, 0.167, 2.288, 0.201, 0.248, 0  'Age': [50, 31, 32, 21, 33, 30, 26, 29, 53, 30],
 'Outcome': [1, 0, 1, 0, 1, 0, 1, 0, 1, 0]
}

diabetes_df = pd.DataFrame(data)

# 1. Display the first five rows
print("First Five Rows:")
print(diabetes_df.head())

# 2. Display the last five rows
print("\nLast Five Rows:")
print(diabetes_df.tail())

# 3. Display dataset information
print("\nDataset Information:")
diabetes_df.info()

# 4. Display the number of rows and columns
print("\nDataset Shape:")
print(diabetes_df.shape)

# 5. Display column names
print("\nColumn Names:")
print(diabetes_df.columns)

# 6. Check for missing values
print("\nMissing Values:")
print(diabetes_df.isnull().sum())

# 7. Display statistical summary
print("\nStatistical Summary:")
print(diabetes_df.describe())

# 8. Check duplicate rows
print("\nNumber of Duplicate Rows:")
print(diabetes_df.duplicated().sum())

# 9. Visualize the correlation between columns
plt.figure(figsize=(10, 6))
sns.heatmap(diabetes_df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title("Correlation Heatmap of Diabetes Dataset")
plt.show()

# 10. Visualize the outcome distribution
sns.countplot(data=diabetes_df, x='Outcome')
plt.title("Diabetes Outcome Distribution")
plt.xlabel("Outcome (0 = No Diabetes, 1 = Diabetes)")
plt.ylabel("Count")
plt.show()
```

```
First Five Rows:
 Pregnancies Glucose BloodPressure SkinThickness Insulin BMI \ 0 6 148
72 35 0 33.6  1 1 85 66 29 0 26.6  2 8 183 64 0 0 23.3  3 1 89 66 23 94
28.1  4 0 137 40 35 168 43.1

 DiabetesPedigreeFunction Age Outcome
0 0.627 50 1
1 0.351 31 0
2 0.672 32 1
3 0.167 21 0
4 2.288 33 1

Last Five Rows:
 Pregnancies Glucose BloodPressure SkinThickness Insulin BMI \ 5 5 116
74 0 0 25.6  6 3 78 50 32 88 31.0  7 10 115 0 0 0 35.3  8 2 197 70 45
543 30.5  9 4 110 92 0 0 37.6
```

```
 DiabetesPedigreeFunction Age Outcome
5 0.201 30 0
6 0.248 26 1
7 0.134 29 0
8 0.158 53 1
9 0.191 30 0

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 9 columns):
 # Column Non-Null Count Dtype
--- ------ -------------- -----
 0 Pregnancies 10 non-null int64
 1 Glucose 10 non-null int64
 2 BloodPressure 10 non-null int64
 3 SkinThickness 10 non-null int64
 4 Insulin 10 non-null int64
 5 BMI 10 non-null float64
 6 DiabetesPedigreeFunction 10 non-null float64
 7 Age 10 non-null int64
 8 Outcome 10 non-null int64
dtypes: float64(2), int64(7)
memory usage: 852.0 bytes

Dataset Shape:
(10, 9)

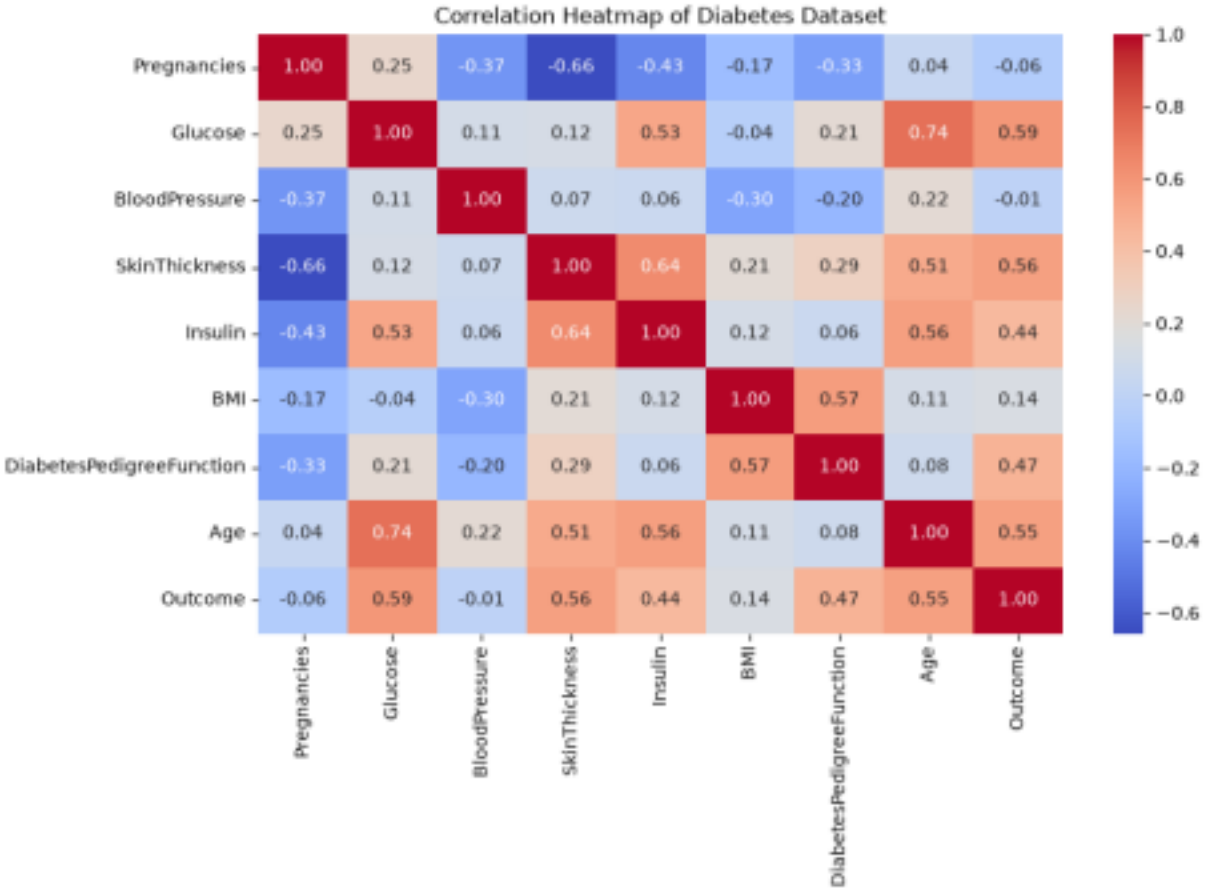
Column Names:
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin',
'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],  dtype='str')
Missing Values:
Pregnancies 0
Glucose 0
BloodPressure 0
SkinThickness 0
Insulin 0
BMI 0
DiabetesPedigreeFunction 0
Age 0
Outcome 0
dtype: int64

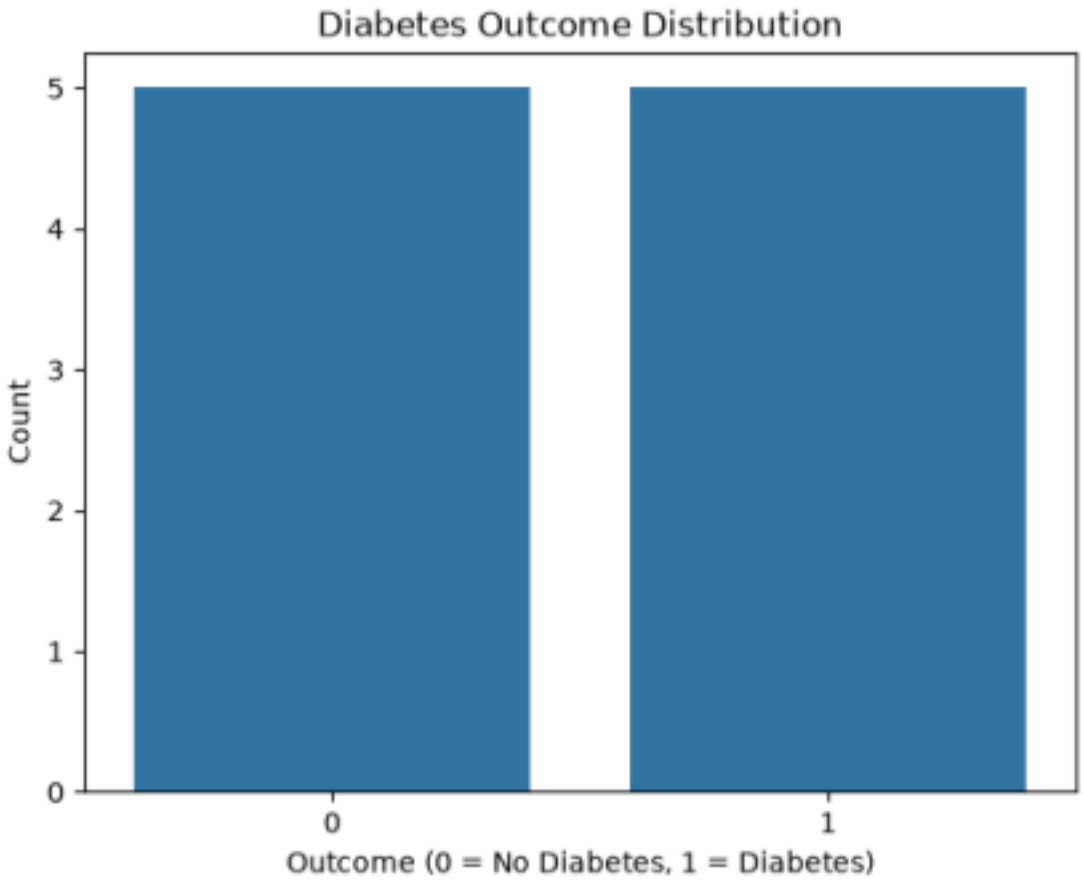
Statistical Summary:
 Pregnancies Glucose BloodPressure SkinThickness Insulin \ count 10.000000
10.000000 10.000000 10.000000 10.000000  mean 4.000000 125.800000 59.400000
19.900000 89.300000  std 3.265986 40.449694 25.051946 17.978073 169.937276
min 0.000000 78.000000 0.000000 0.000000 0.000000  25% 1.250000 94.250000
53.500000 0.000000 0.000000  50% 3.500000 115.500000 66.000000 26.000000
0.000000  75% 5.750000 145.250000 71.500000 34.250000 92.500000  max
10.000000 197.000000 92.000000 45.000000 543.000000

  BMI  DiabetesPedigreeFunction  Age  Outcome    count  10.000000
10.00000  10.000000  10.000000   mean  31.470000  0.50370  33.500000
0.500000  std 6.050537 0.65616 10.102255 0.527046  min 23.300000
0.13400  21.000000  0.000000    25%  26.975000  0.17300  29.250000
0.000000  50% 30.750000 0.22450 30.500000 0.500000  75% 34.875000
0.55800  32.750000  1.000000    max  43.100000  2.28800  53.000000
```

```
1.000000

Number of Duplicate Rows:
0
```





In [ ]: